

VisionInspect AI: Manufacturing Defect Detection & Quality Inspection System

MILESTONE 3

Defect Classification & Manufacturing Analytics

Week 5 & 6

Milestone Progress Document

1. Project Overview

VisionInspect AI: Manufacturing Defect Detection & Quality Inspection System is an AI-powered manufacturing quality-inspection platform intended to detect product defects from images, identify quality issues, classify defect types, and provide production analytics. The platform is designed to reduce manual inspection effort, improve product quality, minimize manufacturing defects, and increase production efficiency.

2. Milestone 3 Objective

Extend the inspection platform from defect detection to defect classification and quality-risk assessment, while building manufacturing analytics for production-quality monitoring and defect-trend analysis.

3. Scope of Work

- Implemented defect categorization workflows.
- Designed severity scoring and quality-risk assessment reports.
- Built quality-assessment and pass/fail decision workflows.
- Generated production-quality reports.
- Built defect analytics dashboards.
- Developed defect-trend monitoring workflows.

4. Planned System Modules

- Defect Classification
- Quality Control
- Manufacturing Analytics Dashboard

5. Key Deliverables

- Defect classification workflow
- Severity scoring framework
- Quality assessment and pass/fail workflow
- Production quality reports
- Manufacturing analytics dashboard
- Defect trend monitoring

6. Severity Scoring Framework

The project specification assigns a severity score using defect size, location, defect type, and detection confidence. The stated weighted framework is:

Parameter	Weight	Purpose
Defect Size	30%	Measures defect size relative to the product surface.

Defect Location	25%	Measures whether the defect is in a critical or non-critical area.
Defect Type	25%	Represents seriousness of the detected defect category.
Detection Confidence	20%	Represents model confidence in the defect prediction.

Severity levels specified in the project document are Critical (80–100), High (60–79), Medium (40–59), and Low (0–39).

7. Expected / Recorded Outcomes

- Defect classification and manufacturing analytics capability.
- Quality-control and reporting workflows.
- Understanding of industrial quality assurance and defect intelligence.
- End-to-end manufacturing inspection workflow from image inspection to quality insight.

8. Technology Stack

Backend: Python (FastAPI) | Frontend: React.js / Next.js | Database: PostgreSQL / MongoDB

AI & Computer Vision: OpenCV, TensorFlow, PyTorch, YOLO, CNN models, Scikit-learn, NumPy and Pandas.

Deployment & Development: Docker, Docker Compose, AWS/Azure, Git/GitHub, VS Code and Postman.

9. Performance & Evaluation Focus

The project specification identifies defect-detection accuracy, precision, recall, F1-score, mAP, inspection automation rate, defect identification accuracy, false-defect detection rate, image-processing speed, inspection response time, and concurrent inspection capability as key metrics.

10. Challenges / Notes

This milestone document is based on the supplied VisionInspect AI project specification. Where implementation evidence, measured model metrics, screenshots, deployment URLs, or issue logs were not included in that source, this document does not invent those details. Actual measured results and screenshots can be added to the GitHub copy before submission.

11. Next Milestone / Continuation

Validate the complete system, optimize model and dashboard performance, deploy the platform, and prepare final technical documentation and presentation.

Conclusion

Milestone 3 establishes the planned progression of the VisionInspect AI system toward an end-to-end manufacturing defect detection and quality inspection platform. The work in this milestone provides the foundation for the next stage of implementation and final demonstration.